DIGITAL CONTROL SYSTEMS

1. a) The equivalent discrete-time model $HP(z)$ is given by

$$\begin{aligned} HP(z) &= (1-z^{-1})Z\left(\frac{1}{s(4s+1)}\right) = (1-z^{-1})Z\left(\frac{1}{s}-\frac{1}{s+\frac{1}{4}}\right) \\ &= (1-z^{-1})\left(\frac{1}{1-z^{-1}}-\frac{1}{1-e^{-\frac{1}{4}}z^{-1}}\right) = \frac{1-e^{-\frac{1}{4}}}{z-e^{-\frac{1}{4}}}. \end{aligned}$$

[4 marks]

b) To discretize the controller $C(s)$ with the Tustin transformation we substitute $s = \dfrac{2}{T}\dfrac{z-1}{z+1}$ with $T=1$, obtaining

$$C_D(z) = C(s)|_{2\frac{z-1}{z+1}} = \frac{(2a+b)z+b-2a}{2(z-1)}.$$

The zero of $C_D(z)$ is $z_0 = \dfrac{2a-b}{2a+b}$. It is easy to see that $|z_0|<1$ for any $a>0$ and any $b>0$, thus concluding that any choice of positive a and b yields a minimium-phase discrete-time controller. [4 marks]

c) According to the sampler locations shown in the figure, it follows that the closed-loop transfer function is

$$W(z) = \frac{Y(z)}{R(z)} = \frac{C_D(z)HP(z)}{1+C_D(z)HP(z)}.$$

which yields

$$\frac{(1-e^{-\frac{1}{4}})((2a+b)z+b-2a)}{2z^2+[(1-e^{-\frac{1}{4}})(2a+b)-2(1+e^{-\frac{1}{4}})]z+2e^{-\frac{1}{4}}+(1-e^{-\frac{1}{4}})(b-2a)}.$$

[4 marks]

d) The easiest way of studying the stability is to use the conditions obtained for second-order polynomials, namely the roots are inside the unit circle if and only if A and B are such that

$$B > -1-A, \qquad B<1, \qquad B>A-1$$

Solving this system of inequalities yields

$$b>0, \qquad b<2(a+1), \qquad a<\frac{1+e^{-\frac{1}{4}}}{1-e^{-\frac{1}{4}}}.$$

Thus, for any a and b such that

$$\begin{cases} 0<a<\dfrac{1+e^{-\frac{1}{4}}}{1-e^{-\frac{1}{4}}} \approx 8.042, \\ 0<b<2(1+a), \end{cases}$$

the closed-loop system is asymptotically stable and the controller is minimum-phase. This last property follows from part b) where we found that the controller is minimum phase for any $a>0$, $b>0$. [8 marks]

Answers

2. a) The relation between the matrices (A,B,C) and (F,G,L) are the following

$$A = e^{FT}, \qquad B = \int_0^T e^{F\lambda} G d\lambda, \qquad C = L.$$

To compute these matrices we first compute $(sI-F)^{-1}$, namely

$$(sI-F)^{-1} = \begin{bmatrix} \frac{1}{s} & \frac{1}{s(s+1)} \\ 0 & \frac{1}{s+1} \end{bmatrix}.$$

It follows that

$$A = \begin{bmatrix} 1 & 1-e^{-1} \\ 0 & e^{-1} \end{bmatrix}, \qquad B = \begin{bmatrix} e^{-1} \\ 1-e^{-1} \end{bmatrix}.$$

[4 marks]

b) To compute the transfer function we first determine the term

$$(zI-A)^{-1} = \begin{bmatrix} \frac{1}{z-1} & \frac{1-e^{-1}}{(z-1)(z-e^{-1})} \\ 0 & \frac{1}{z-e^{-1}} \end{bmatrix}.$$

The input-output transfer function is given by

$$\frac{Y(z)}{U(z)} = C(zI-A)^{-1}B = \frac{e^{-1}z+1-2e^{-1}}{(z-1)(z-e^{-1})}.$$

[3 marks]

c) The equivalent discrete-time model is

$$\begin{aligned} HP(z) &= \frac{z-1}{z} Z\left(\frac{P(s)}{s}\right) = \frac{z-1}{z} Z\left(\frac{1}{s+1} - \frac{1}{s} + \frac{1}{s^2}\right) \\ &= \frac{z-1}{z}\left(\frac{1}{1-e^{-1}z^{-1}} - \frac{1}{1-z^{-1}} + \frac{Tz^{-1}}{(1-z^{-1})^2}\right) \\ &= \frac{e^{-1}z+1-2e^{-1}}{(z-1)(z-e^{-1})}. \end{aligned}$$

as in b).

[3 marks]

d) Selecting $u(k) = -Kx(k)$ we obtain the closed-loop state equation

$$x(k+1) = (A-BK)x(k).$$

To achieve a deadbeat response we place the eigenvalues of $(A-BK)$ in zero. Let $K = \begin{bmatrix} K_1 & K_2 \end{bmatrix}$. Then,

$$\begin{aligned} \det(\lambda I - (A-BK)) = \lambda^2 &+ (e^{-1}K_1 + (1-e^{-1})K_2 - 1 - e^{-1})\lambda \\ &+ (1-2e^{-1})K_1 + (e^{-1}-1)K_2 + e^{-1}. \end{aligned}$$

To place the eigenvalues in zero we solve the equations

$$\begin{aligned} e^{-1}K_1 + (1-e^{-1})K_2 &= 1+e^{-1}, \\ (1-2e^{-1})K_1 + (e^{-1}-1)K_2 &= -e^{-1}, \end{aligned}$$

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which gives

$$K_1 = \frac{1}{1-e^{-1}} \approx 1.582, \qquad K_2 = \frac{1-e^{-1}-e^{-2}}{(1-e^{-1})^2} \approx 1.243.$$

For completeness, although not required, the closed-look system is

$$\begin{aligned} x(k+1) = (A-BK)x(k) &= \begin{bmatrix} \frac{1-2e^{-1}}{1-e^{-1}} & \left(\frac{1-2e^{-1}}{1-e^{-1}}\right)^2 \\ -1 & -\frac{1-2e^{-1}}{1-e^{-1}} \end{bmatrix} x(k) \\ &\approx \begin{bmatrix} 0.418 & 0.175 \\ -1 & -0.418 \end{bmatrix} x(k). \end{aligned}$$

[4 marks]

e) Note that for any $(x_1(0), x_2(0))$ we have

$$\begin{aligned} x_1(1) &= 0.418x_1(0) + 0.175x_2(0), \\ x_2(1) &= -x_1(0) - 0.418x_2(0), \end{aligned}$$

and

$$\begin{aligned} x_1(2) &= 0.418x_1(1) + 0.175x_2(1) \\ &= 0.418(0.418x_1(0) + 0.175x_2(0)) + 0.175(-x_1(0) - 0.418x_2(0)) \\ &= 0.175x_1(0) + 0.073x_2(0) - 0.175x_1(0) - 0.073x_2(0) = 0, \end{aligned}$$

$$\begin{aligned} x_2(2) &= -x_1(1) - 0.418x_2(1) \\ &= -(0.418x_1(0) + 0.175x_2(0)) - 0.418(-x_1(0) - 0.418x_2(0)) \\ &= -0.418x_1(0) - 0.175x_2(0) + 0.418x_1(0) + 0.175x_2(0) = 0. \end{aligned}$$

This is expected as a deadbeat controller produces finite-time convergence to zero in at most n iterations. [3 marks]

f) The closed-loop transfer function of the block diagram is given by

$$\frac{Y(z)}{R(z)} = \frac{C(z)HP(z)}{1+C(z)HP(z)}.$$

Thus, the characteristic polynomial is

$$\text{num}(1+C(z)HP(z)).$$

The problem has a unique solution if we select the control $C(z)$ as

$$C(z) = \frac{s_0 z + s_1}{z + v_1},$$

because $m = \deg(\text{den}(HP(z))) - 1$. Thus,

$$\begin{aligned} &\text{num}(1+C(z)HP(z)) = z^3 + (v_1 - (1+e^{-1}) + s_0 e^{-1})z^2 + \\ &+(-v_1(1+e^{-1}) + e^{-1} + e^{-1}s_1 + (1-2e^{-1})s_0)z + (e^{-1}v_1 + (1-2e^{-1})s_1). \end{aligned}$$

To achieve a deadbeat response, we require that $\text{num}(1+C(z)HP(z)) = z^3$, which yields

$$\begin{bmatrix} e^{-1} & 0 & 1 \\ 1-2e^{-1} & e^{-1} & -1-e^{-1} \\ 0 & 1-2e^{-1} & e^{-1} \end{bmatrix} \begin{bmatrix} s_0 \\ s_1 \\ v_1 \end{bmatrix} = \begin{bmatrix} 1+e^{-1} \\ -e^{-1} \\ 0 \end{bmatrix}.$$

For the sake of completeness the solution of this linear system is $s_0 = 2.3055$, $s_1 = -0.7236$, $v_1 = 0.5197$. However, it was not required to determine these values. Note that $K_1 = s_0 + s_1$ and $K_2 = v_1 - s_1$. [3 marks]

Answers

3. a) i) Consider

$$P(s) = \frac{\frac{1}{2}}{(s^2+\frac{1}{4})(s^2+\frac{9}{4})}.$$

The poles are the roots of the denominator, that is

$$p_1 = \frac{1}{2}j, \quad p_2 = -\frac{1}{2}j, \quad p_3 = \frac{3}{2}j, \quad p_4 = -\frac{3}{2}j.$$

Since the real part of all these poles is zero and the poles are distinct, it can be concluded that the continuous-time system described by $P(s)$ is marginally stable. [3 marks]

ii) For the generic sampling time T, the discrete-time equivalent is given by

$$\begin{aligned} HP(z) &= (1-z^{-1})Z\left(\frac{\frac{1}{2}}{(s^2+\frac{1}{4})(s^2+\frac{9}{4})}\right) \\ &= (1-z^{-1})\left(\frac{c_1}{s} + \frac{c_2}{s-\frac{1}{2}j} + \frac{c_2^*}{s+\frac{1}{2}j} + \frac{c_3}{s-\frac{3}{2}j} + \frac{c_3^*}{s+\frac{3}{2}j}\right). \end{aligned}$$

Simple algebra shows that $c_1 = \frac{8}{9}$, $c_2 = c_2^* = -\frac{1}{2}$, $c_3 = c_3^* = \frac{1}{18}$. Thus,

$$\begin{aligned} HP(z) = (1-z^{-1})Z\Bigg(&\frac{\frac{8}{9}}{1-z^{-1}} - \frac{1}{2}\left(\frac{1}{1-e^{-j\frac{T}{2}}z^{-1}} + \frac{1}{1-e^{j\frac{T}{2}}z^{-1}}\right) \\ &+ \frac{1}{18}\left(\frac{1}{1-e^{-j\frac{3T}{2}}z^{-1}} + \frac{1}{1-e^{j\frac{3T}{2}}z^{-1}}\right)\Bigg). \end{aligned}$$

Finally, we obtain the required form

$$HP(z) = \frac{8}{9} + (z-1)\left(\frac{\cos(T/2)-z}{z^2-2z\cos(T/2)+1} + \frac{1}{9}\frac{z-\cos(3T/2)}{z^2-2z\cos(3T/2)+1}\right).$$

[4 marks]

iii) Substituting $T = \pi$ sec in the expression of $HP(z)$ obtained in part ii) yields

$$HP(z) = \frac{8}{9}\frac{z+1}{z^2+1}.$$

This sampling time is not implementable in practice because π is irrational but a digital machine can represent only a finite number of digits. [3 marks]

iv) For a given value of the sampling time T, a point s in the Laplace complex plane is mapped into a point z in the Z-plane through the relationship

$$z = e^{sT}.$$

Applying this relationship to the poles of $G(s)$ computed in part i) with $T = \pi$ sec yields

$$\begin{aligned} z_1 = e^{p_1 T} = e^{j\frac{\pi}{2}} = j, \qquad z_2 = e^{p_2 T} = e^{-j\frac{\pi}{2}} = -j, \\ z_3 = e^{p_3 T} = e^{j\frac{3\pi}{2}} = -j, \quad z_4 = e^{p_4 T} = e^{-j\frac{3\pi}{2}} = j. \end{aligned}$$

Thus, the four continuous-time poles are mapped in only two distinct points in the Z-plane. This is caused by the specific choice of T for

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which only poles p_1 and p_2 fall in the primary strip $\left(-\frac{\omega_s}{2}, \frac{\omega_s}{2}\right)$ with $\omega_s = \frac{2\pi}{T} = 2$ rad/sec and for the fact that the pair p_3 and p_4 lie in the complementary strip at a distance from the pair p_1 and p_2 that is exactly equal to the sampling time. Hence, this choice of the sampling time causes an aliasing phenomenon. [4 marks]

b) i) Note that $X(z)$ can be rewritten as

$$X(z) = \frac{z(z-2)}{(z-1)^2}x(0) + \frac{z}{(z-1)(z-\frac{1}{2})^2}.$$

Computing the partial fraction expansion of $\frac{X(z)}{z}$ we obtain

$$X(z) = \frac{z}{z-1}x(0) - \frac{z}{(z-1)^2}x(0) + 4\frac{z}{z-1} - 4\frac{z}{z-\frac{1}{2}} - 2\frac{z}{(z-\frac{1}{2})^2}.$$

The inverse z-transform of this last expression is

$$x(k) = (1-k)x(0) + 4 - 4(1+k)\left(\frac{1}{2}\right)^k.$$

[3 marks]

ii) Note that

$$\lim_{k\to\infty}(1+k)\left(\frac{1}{2}\right)^k \to 0,$$

Hence, $x(k)$ behaves as $(1-k)x(0)+4$. If $x(0) = 0$ the solution $x(k)$ converges asymptotically to 4. If $x(0) > 0$, the solution $x(k)$ diverges to $-\infty$. If $x(0) < 0$, the solution $x(k)$ diverges to $+\infty$.

[3 marks]

Answers